

Spacetime Engineering

An Introduction

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Preface

Could we build a shortcut through space? Could a ship cross the galaxy while its passengers age only a few years, or a tunnel connect two stars? General relativity gives these questions a precise form. It tells us that space and time together make up a geometry that matter can bend, and it gives the rule connecting the two. Once that rule is in hand, the question stops being “is this allowed by physics?” in the abstract. It becomes an engineering question: what geometry would do the job, what would it take to produce that geometry, and can nature supply what it takes?

This book teaches you to work through that question. By the end of it you will be able to write down a spacetime metric designed to do something, work out the matter and energy it requires, test that requirement against everything physics knows about matter, and judge how far the design is from anything that could be built. Along the way you will meet the classic designs of the field: traversable wormholes, warp drives and Krasnikov tubes, together with the results that constrain them and the questions that remain open.

Who this book is for

The book is written for upper-level undergraduates in physics, graduate students in engineering, and anyone with similar preparation. You should be comfortable with special relativity at the level of an introductory modern physics course, with multivariable calculus and linear algebra, and with solving ordinary differential equations. You do not need to have studied general relativity: the first chapters develop the parts of it this book uses, starting from spacetime diagrams. Readers who already know general relativity can move quickly through those chapters and will find the material aimed at them in the boxes marked *Going deeper*.

How the book is organized

Part I builds the foundations: what spacetime engineering is, the geometry of spacetime, the way matter shapes that geometry, and the physical limits on matter that decide what can be built. Part II studies the classic designs one at a time. Part III turns to design itself: how each part of a spacetime’s geometry affects what it demands and what it does to travellers.

Part IV asks what kinds of matter and fields could supply those demands. Part V takes up dynamics, the checking of calculations, and the open problems of the field.

How to use this book

Each chapter opens with a short list of what you will learn and closes with a summary. Worked examples show each method carried through in full, with the reasoning spelled out. The *Check your understanding* questions in the text are short; try each one before reading on, and compare with the answers at the end of the chapter. The problems at the end of each chapter are marked by difficulty: [★] problems practise a single idea, [★★] problems combine several, and [★★★] problems are open-ended or demanding. Boxes marked *Going deeper* carry material for readers with more background, and can be skipped on a first reading.

Part I

Foundations

Chapter 1

What Is Spacetime Engineering?

What you will learn

- What it means to engineer spacetime, and how that differs from engineering inside a gravitational field that nature has already fixed.
- The two directions in which Einstein's equation can be read, and why a spacetime engineer reads it "backwards".
- The landmark designs of the field (traversable wormholes, warp drives and Krasnikov tubes) and what each one is meant to do.
- Why these designs are hard to build: the unusual matter they need, the limits quantum physics places on that matter, and the threat they pose to cause and effect.
- How the rest of the book is organized.

This chapter is a tour. It introduces the ideas of the whole subject with as little mathematics as possible, so that you know where the later chapters are heading and why each one matters. Every claim made here in words is made again later in equations, and most of them you will derive yourself.

1.1 You already depend on engineered time

Every time a phone finds its position by satellite, it trusts clocks that are orbiting about 20 000 km overhead. The navigation satellites broadcast the time on their onboard atomic clocks, and your receiver works out how far away each satellite is from how long its signal took to arrive. Light travels about 30 cm in a nanosecond, so the clocks must agree with one another to within a few tens of nanoseconds for the position to be good to a few metres.

Here is the remarkable part. Two identical atomic clocks, one on the ground and one in a navigation satellite, do not tick at the same rate. The satellite clock gains about 45 microseconds per day because it sits higher in Earth's gravity, where time runs faster. It

loses about 7 microseconds per day because it moves at almost 4 km/s, and moving clocks run slow. The net effect is that the orbiting clock runs fast by roughly 38 microseconds every day. Left uncorrected, that offset would produce a ranging error of about 11 km after a single day, since light covers that distance in 38 microseconds. The system works because its designers account for both effects (Ashby 2003). You will calculate both numbers yourself in Chapter 2.

Navigation engineers are working *within* a geometry that nature has fixed. The rates at which clocks tick at different altitudes are set by the mass of the Earth, and the engineers' job is to measure those rates and correct for them. Spacetime engineering asks the reverse question. Suppose we could choose the geometry. What pattern of clock rates, distances and light paths would we want? What would such a spacetime do to travellers, to light, and to the ordinary matter around it? And what would it take to produce it?

1.2 Spacetime has a geometry

Special relativity teaches that space and time are not separate. Observers in relative motion disagree about lengths and durations. They agree, however, on a combination of the two called the *interval* between events, and on the speed of light. Together, space and time form a four-dimensional *spacetime*, and the interval plays the role that distance plays in ordinary geometry.

General relativity takes one further step. It says that spacetime is not a fixed, flat stage. Its geometry can be curved, and gravity *is* that curvature. An apple falls because it follows the straightest possible path through a curved spacetime, and clocks at different heights tick at different rates because the geometry of time itself varies from place to place. The physicist John Wheeler summarized the theory in one sentence: spacetime tells matter how to move, and matter tells spacetime how to curve.

The rule for the second half of that sentence is *Einstein's equation* (Einstein 1916). In words it says

$$(\text{curvature of spacetime}) = \frac{8\pi G}{c^4} (\text{energy, momentum, pressure and stress of matter}). \quad (1.1)$$

The left side is built from the geometry. The right side describes everything that has energy: matter, light, fields, and even the pressure and tension inside a material. Chapter 3 turns both sides into precise mathematical objects, and by the end of that chapter you will be able to use the equation yourself.

The constant in front deserves attention right away, because it sets the scale of the entire subject. In SI units,

$$\frac{c^4}{G} \approx 1.2 \times 10^{44} \text{ N}. \quad (1.2)$$

This is a force, and an enormous one. Roughly speaking, bending spacetime so that it curves appreciably over a distance L requires stresses of order $c^4/(GL^2)$. For $L = 1$ m that is about 10^{44} Pa, more than thirty orders of magnitude beyond the strength of the strongest materials we know. Spacetime is extraordinarily stiff. This single number explains why the geometry around you is so nearly flat, and it is the first reason spacetime engineering is hard.

Check your understanding 1.1

The Earth's gravity bends spacetime enough to make clocks on the ground and in orbit disagree measurably. How can that be, if spacetime is so stiff? What property of the Earth makes up for the huge constant in Equation (1.2)?

1.3 Reading Einstein's equation in both directions

Physicists usually read Einstein's equation from right to left. They describe the matter first (a star, a galaxy, the contents of the universe) and then solve the equation for the geometry that matter produces. This is how the geometry around the Sun was found, and the geometry of black holes, and the expanding universe of cosmology. It is also hard: the equation is a set of coupled, nonlinear partial differential equations, and only highly symmetric problems can be solved by hand.

A spacetime engineer can read the equation from left to right instead. Start by deciding what you want the spacetime to do: carry a traveller quickly between two points, keep the traveller's clock in step with clocks at home, protect the traveller from tidal forces. Write down a geometry that does those things. Then compute its curvature, and read off from Equation (1.1) the matter that must be present to produce it. We will call that matter the *demand*ed matter of the geometry. In this direction the mathematics is easy: computing curvature from a given geometry takes only differentiation.

Michael Morris and Kip Thorne introduced this way of working in 1988, in a paper written for students (Morris and Thorne 1988). The literature now calls it *metric engineering*, because the geometry is specified by its metric, the object you will meet in Chapter 2 (McMonigal et al. 2012; Schuster et al. 2023). Morris and Thorne designed spacetimes containing traversable wormholes by first listing the properties a traveller would need, then choosing a geometry with those properties, and only then asking what matter the geometry would require.

The catch comes at the last step. Einstein's equation always returns an answer: *every* geometry has some demanded matter. Nothing guarantees that such matter exists. The demand might call for energy densities that are negative, or tensions larger than any material can bear, or both. The real work of spacetime engineering begins there, and it proceeds in a loop (Figure 1.1).

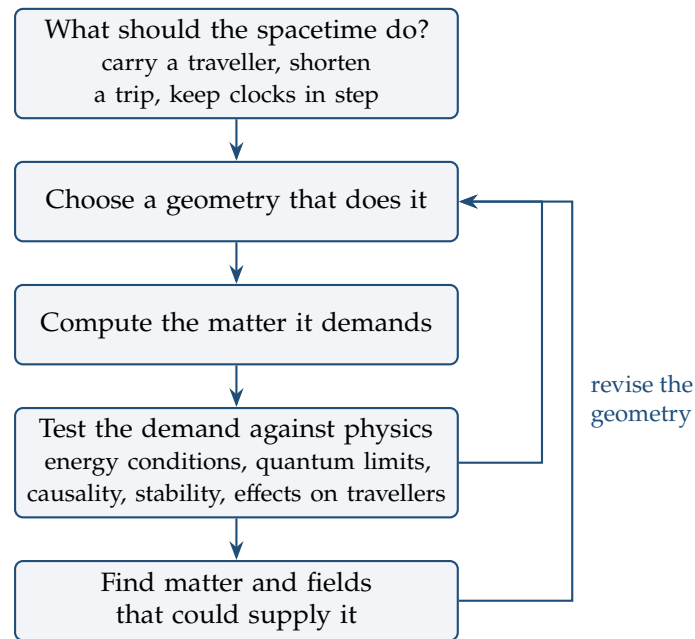


Figure 1.1: The design loop of spacetime engineering. A geometry is chosen for what it does, and its demanded matter is tested against physics. When no known matter can supply the demand, the geometry is revised. Most of this book teaches the steps inside this loop.

The loop can also run the other way. Some recent work starts from a matter distribution with sensible properties, such as a shell of ordinary matter, solves for the geometry it produces, and then asks what that geometry can do. Both directions appear in this book. The geometry-first direction is the more common one, and it is the one that most clearly exposes what a design costs.

Check your understanding 1.2

Why is it always possible to compute the demanded matter of a geometry, while solving for the geometry produced by given matter is often impossible to do by hand?

1.4 A gallery of engineered spacetimes

Three designs define the field. Each was invented to answer the question “could we travel between the stars much faster than light takes?”, and each answers it in a different way.

1.4.1 Traversable wormholes

A wormhole is a tunnel through space: two openings, perhaps light-years apart, joined by a short passage. Figure 1.2 shows the usual way of picturing one. Take a two-dimensional slice through space around the wormhole, and draw it as a curved surface. Far from the wormhole the surface is flat. Near it, the surface curves down into a funnel, narrows to a smallest circle called the *throat*, and opens out again into a second region of space.

Wormholes appear as mathematical solutions in general relativity going back to the 1930s, but the early examples could not be crossed. Morris and Thorne asked what a wormhole would need in order to be *traversable*: no horizon at the throat, tidal forces a person could survive, and a crossing that takes a reasonable time. They found geometries that meet all these requirements, and then computed what those geometries demand.

Historical note: a novel and a wormhole

Morris and Thorne were prompted to study traversable wormholes in the summer of 1985, when the astronomer Carl Sagan sent Thorne a draft of his novel *Contact* and asked for help making its gravitational physics as accurate as possible (Morris and Thorne 1988). Sagan revised the novel’s travel method in light of their preliminary results. The resulting paper, subtitled “a tool for teaching general relativity”, launched the modern study of engineered spacetimes.

The difficulty lies at the throat. A bundle of light rays heading into a wormhole converges as it approaches the throat. To come out the other side, the bundle must spread apart again.

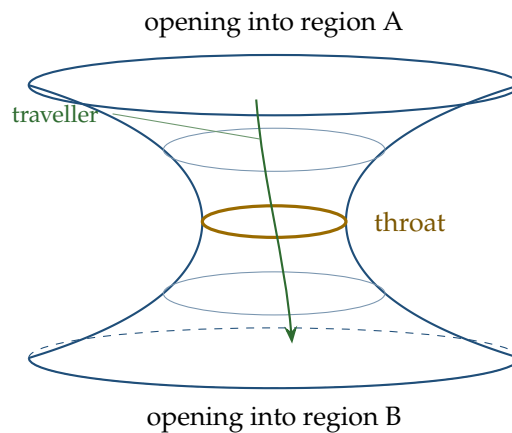


Figure 1.2: An embedding diagram of a wormhole. A two-dimensional slice of space is drawn as a surface. Circles around the wormhole shrink toward the throat and grow again on the far side: the throat *flares out* in both directions.

Ordinary matter, however, always makes light rays converge: gravity attracts, and it bends light the way a converging lens does. A throat has to act as a *diverging* lens. Morris and Thorne showed that this requires the material at the throat to carry a tension greater than its energy density. Observers passing through the throat at speeds close to that of light would then measure a *negative* energy density. Matter with this property is called *exotic*. You will derive the flaring-out condition and its consequence for yourself in Part II.

1.4.2 Warp drives

In 1994 Miguel Alcubierre took a different approach (Alcubierre 1994). Instead of connecting two distant places by a tunnel, his design moves a region of space itself. Space contracts in front of a bubble and expands behind it, and a region of perfectly flat spacetime inside the bubble is carried along (Figure 1.3). A ship inside that region does not move through its surroundings at all; it floats in free fall while the bubble carries it.

Alcubierre's bubble has several striking properties. Nothing inside it ever moves faster than light relative to its own surroundings, yet the bubble as a whole can move at any speed relative to distant observers. The ship feels no acceleration. In Alcubierre's design its clock ticks at the same rate as clocks far away, so the travellers age no more than the people they left behind during the trip. José Natário later showed that the expansion and contraction are not essential: a warp drive can carry its passengers with no change in volume at all (Natário 2002). What *is* essential is the demand. The bubble's wall requires negative energy density, just as the wormhole's throat does.

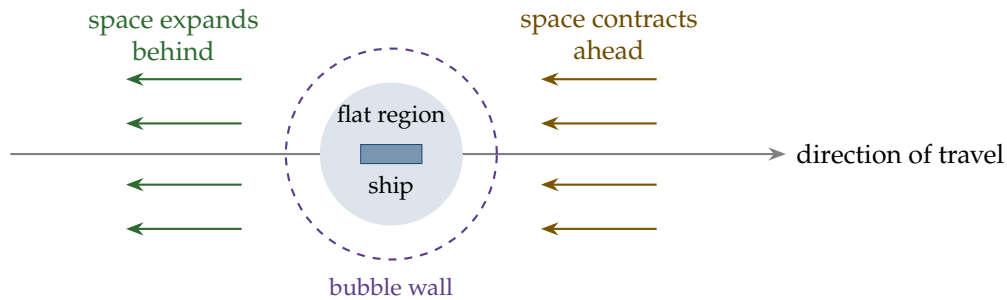


Figure 1.3: A schematic warp drive. Space contracts in front of the bubble and expands behind it, and the flat region inside is carried along. The ship inside is in free fall.

1.4.3 Krasnikov tubes

Sergei Krasnikov pointed out a problem with the warp drive as a means of travel: the people inside a bubble moving faster than light cannot send signals to its front edge, so they cannot create the bubble or steer it (Krasnikov 1998). His alternative has the traveller build the structure on the way out. A ship travels to a distant star at less than light speed, and as it goes it modifies the spacetime along its path, leaving behind a tube. The modified geometry inside the tube lets the return trip arrive home shortly after departure. Allen Everett and Thomas Roman analysed these tubes further and showed that two of them, placed suitably, could form a time machine (A. E. Everett and T. A. Roman 1997).

1.4.4 Recent work

In 2021 several authors proposed warp drives that appeared to require only positive energy. Examining them closely, Jessica Santiago, Sebastian Schuster and Matt Visser showed that the positive energy was measured by one special family of observers only. Other observers see negative energy, and every physically reasonable warp drive violates the conditions that ordinary matter satisfies (Santiago et al. 2022). That episode taught the field a lesson you will meet again and again in this book: the demand of a geometry must be checked from the point of view of *every* observer. Current research includes shells of ordinary matter that carry a flat interior at speeds below that of light, computer simulations of warp spacetimes, and the search for fields that could supply negative energy.

1.5 The central obstacle

All the designs in the gallery share one requirement. Somewhere, they need matter that bends light the “wrong” way.

1.5.1 Energy conditions

Physicists summarize what ordinary matter can do with a small set of inequalities called *energy conditions*. The simplest to state is the *weak energy condition*: every observer, however they move, measures a non-negative energy density. The most important for this book is the *null energy condition*, which is weaker still. It concerns light rays passing through matter, and it guarantees that matter always focuses a bundle of light rays rather than spreading it. Everything you have ever touched satisfies these conditions, and so do light, electric and magnetic fields, and all ordinary fluids.

The null energy condition is exactly what shortcuts need to violate. A wormhole throat must defocus light, and a warp bubble's wall must do the same. In Chapter 5 you will see that this is no accident: under quite general assumptions, a spacetime that lets travellers outrun light through an otherwise ordinary region of space must violate the null energy condition somewhere.

Key idea

Designing a shortcut through spacetime is easy on paper. Supplying the matter it demands is the hard part, because shortcuts demand matter that violates the null energy condition, and no ordinary matter does.

1.5.2 Negative energy in the laboratory

Negative energy density is not pure fiction. In 1948 Hendrik Casimir predicted that two parallel, uncharged metal plates in vacuum attract each other (Casimir 1948). The plates restrict which electromagnetic vacuum fluctuations fit between them, and the energy density between the plates is lower than that of empty space: in this sense it is negative. Steve Lamoreaux measured the Casimir force in 1997 (Lamoreaux 1997), and it has since been measured many times. Quantum physics therefore allows negative energy.

It allows it only in limited amounts. Lawrence Ford and Thomas Roman showed that quantum fields obey *quantum inequalities*: the more negative the energy density in a region, the shorter the time it can last and the smaller the region it can occupy (Ford and T. A. Roman 1995, 1997). Applied to a warp bubble, these inequalities force the bubble's wall to be extraordinarily thin. Michael Pfenning and Ford estimated that a bubble with a radius of 100 m would then need about 6×10^{62} kg of negative energy for every unit of its speed in units of c (Pfenning and Ford 1997). That is vastly more than the total mass of the observable universe. Clever reshaping of the bubble can cut the requirement dramatically: Chris Van Den Broeck brought it down to a few solar masses (Van Den Broeck 1999). That is still an astronomical amount, and his design bends space on scales of about ten Planck lengths, far below anything that has ever been probed.

Check your understanding 1.3

The Casimir effect shows that negative energy density exists. Why does that not settle the question of whether a warp drive can be built?

1.5.3 Causality and control

Two further obstacles have nothing to do with the amount of energy. The first is causality. Relativity allows observers in relative motion to disagree about which of two distant events happens first. If a signal or a traveller could outrun light, some observers would see it arrive before it left. With two faster-than-light structures, a traveller could return before departing and change the past. Stephen Hawking conjectured that physics always prevents this, a proposal he called the *chronology protection conjecture* (Hawking 1992). It remains a conjecture, and any practical design must show that it creates no such loops.

The second is control. As Krasnikov noticed, the occupants of a warp bubble moving faster than light are cut off from the bubble's front edge. They can neither build the bubble nor steer it from inside. Designs that carry travellers faster than light must therefore be laid out in advance, or driven by something outside the travellers' reach.

1.6 What spacetime engineering is today

With all these obstacles, what is the subject for? Nobody knows how to build any of the spacetimes in this book, and some of them may be forbidden outright. Spacetime engineering today is a theoretical discipline, and it earns its place in three ways.

It teaches general relativity. Designing a spacetime forces you to understand how every piece of its geometry affects clocks, rulers, light and matter. That was Morris and Thorne's reason for writing their paper, and it is one reason for this book.

It maps the boundary of the possible. Some of the obstacles above are theorems with precise assumptions. Some are estimates that depend on how a design is drawn. Some are conjectures. Knowing which is which tells you what would have to change for a design to become possible, and a large part of this book is about learning to tell them apart.

It develops methods that carry over to other problems: how to design under severe physical constraints, how to check a long calculation, and how to judge how far a result can be trusted. Those methods are useful long before anyone builds a warp drive.

1.7 How this book is organized

The rest of Part I gives you the tools. Chapter 2 develops the geometry of spacetime: intervals, proper time, the metric, curvature and free fall. Chapter 3 describes matter by its

energy, momentum and stress, states Einstein's equation, and shows how to compute the matter a geometry demands. The chapters that follow split spacetime into space and time the way an engineer describes a design, state the energy conditions precisely, explain what quantum physics allows, and treat causality and the shape of space as a whole.

Part II studies the classic designs in detail. Part III is about design: how the rate of clocks, the flow of space and the shape of space each affect what a spacetime demands and what it does to the people inside it. Part IV asks what kinds of matter and fields could supply a demand. Part V takes up dynamics, the checking of calculations, and the open problems of the field.

In this chapter we have used SI units. From Chapter 2 onward we measure time and distance in the same units, which sets $c = 1$, and from Chapter 3 onward we also set $G = 1$. Each chapter shows how to convert back to everyday units whenever a number matters.

Summary

- Spacetime has a geometry that matter bends; gravity is that bending. Einstein's equation links the curvature of spacetime to the energy, momentum, pressure and stress of matter.
- The constant $c^4/G \approx 1.2 \times 10^{44}$ N makes spacetime extremely stiff. Large changes in geometry require enormous amounts of energy and stress.
- Spacetime engineering reads Einstein's equation backwards: choose a geometry for what it does, then compute the matter it demands. Every geometry has a demanded matter content; whether that matter exists is the real question.
- The classic designs are traversable wormholes (tunnels through space), warp drives (bubbles of flat space carried along by a flow of space) and Krasnikov tubes (structures laid down on an outbound trip).
- All of them demand matter that violates the null energy condition, which every ordinary material satisfies. Quantum physics allows negative energy density, as the Casimir effect shows, but quantum inequalities limit how much and for how long.
- Faster-than-light designs also face causality (the risk of time machines) and control (occupants cannot steer a faster-than-light bubble from inside).

Answers to Check your understanding

- 1.1 The stiffness constant multiplies the *density* of energy. The Earth's mass, about 6×10^{24} kg, is concentrated enough that its gravitational potential at the surface is about 7×10^{-10} in units of c^2 . That tiny fraction is exactly the size of the clock-rate

differences GPS must correct. Large masses produce small curvature, and precise clocks can measure small curvature.

- 1.2 Computing the demanded matter only requires differentiating the geometry, which can always be done. Solving for the geometry means solving nonlinear partial differential equations for unknown functions, which is possible by hand only in highly symmetric cases.
- 1.3 The quantum inequalities limit how large and long-lived a region of negative energy can be. A warp drive needs large amounts of negative energy held in place for the duration of the trip, far beyond what the inequalities allow for known quantum fields.

Problems

- 1.1 [★] **Two clocks.** A clock sits at the top of a tower of height $h = 100$ m. Near the Earth's surface, a clock higher by h runs faster by the fraction gh/c^2 , where $g = 9.8$ m/s². How many nanoseconds does the top clock gain over a clock at the base in one year?
- 1.2 [★] **The stiffness of spacetime.** Compute c^4/G in newtons from $c = 2.998 \times 10^8$ m/s and $G = 6.674 \times 10^{-11}$ N m²/kg². Then estimate the stress, in pascals, needed to curve spacetime appreciably over a distance of 1 km, and compare it with the tensile strength of steel, about 10^9 Pa.
- 1.3 [★★] **A day of GPS error.** Using the numbers in Section 1.1, find how long it takes for an uncorrected satellite clock to drift by the 30 ns that corresponds to about 10 m of position error.
- 1.4 [★★] **Converging and diverging lenses.** Explain in your own words why a wormhole throat must defocus light, and why ordinary matter cannot do that. (Hint: think about how a bundle of light rays behaves as it enters the throat, passes through, and leaves on the other side.)
- 1.5 [★★] **How much is too much?** Express the Pfenning–Ford estimate of 6×10^{62} kg in solar masses, taking $M_\odot = 2.0 \times 10^{30}$ kg. Van Den Broeck's design needs a few solar masses. By what factor did reshaping the bubble reduce the requirement?
- 1.6 [★ ★ ★] **Which assumptions?** The quantum inequalities rest on assumptions: the kind of quantum field, the flatness of spacetime over the sampling time, and the way the energy is averaged. For each, describe how a designer might try to take advantage of it, and what you would check to see whether the attempt succeeds. Return to your answer after Chapter 6.

Further reading

Morris and Thorne (1988)

The paper that started the field, written for students and still one of the best introductions to engineered spacetimes.

Alcubierre (1994)

A five-page paper introducing the warp drive, readable after Chapter 3.

A. Everett and T. Roman (2011)

A book-length, non-technical account of warp drives, wormholes and time travel by two researchers who helped shape the field.

Visser (1995)

The standard monograph on wormholes, for readers who want the full mathematical treatment.

Kontou and Sanders (2020)

A modern review of energy conditions in classical and quantum physics, useful as a reference throughout this book.

Ashby (2003)

The complete relativistic treatment of the Global Positioning System.

Chapter 2

The Geometry of Spacetime

What you will learn

- Draw and read spacetime diagrams, worldlines and light cones.
- Compute the interval between two events and say whether it is timelike, spacelike or null.
- Compute the time a clock records along any path through spacetime.
- Use a metric to measure intervals, and tell a coordinate effect from a real feature of the geometry.
- Compute how fast clocks tick in a weak gravitational field and near a black hole.
- Describe free fall as motion along a geodesic, and curvature as the source of tidal forces.

A spacetime engineer designs geometries, so we need a precise language for geometry. This chapter builds it in steps. We start with the flat spacetime of special relativity and learn to measure it with the interval. We then write the interval in a form, the *metric*, that works in any coordinates and in curved spacetime too. With the metric in hand, we can compute everything a traveller would experience: the ticking of clocks, the paths of light, the motion of falling objects, and the stretching forces called tides.

2.1 Events, worldlines and units

An *event* is a place at an instant: a firecracker going off at a particular point and time. We label events with four coordinates, a time t and three positions x, y, z . A particle traces out a sequence of events called its *worldline*.

A *spacetime diagram* plots worldlines with time running up the page and one direction of space across it (Figure 2.1). A particle at rest has a vertical worldline. A moving particle has a tilted one, and the faster it moves the more its worldline leans over.

It pays to measure time and distance in the same units. A light-second is the distance light travels in one second, about 3×10^8 m. If we measure distance in light-seconds and time in seconds, or equivalently time in metres (one metre of time is the time light takes to travel a metre, about 3.3 ns), then the speed of light is exactly $c = 1$. Velocities become pure numbers: a speed of 0.5 means half the speed of light. From here on we use these units, and restore factors of c when we need a number in everyday units. Light worldlines then run at 45° on every spacetime diagram.

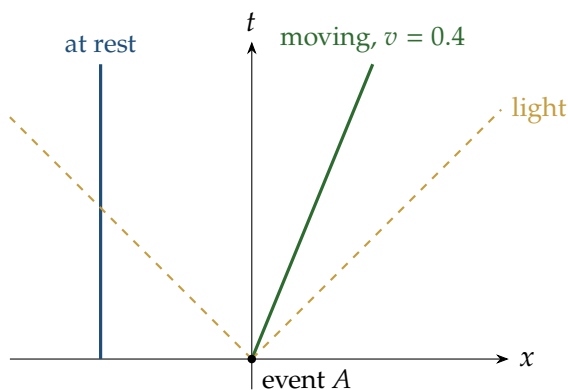


Figure 2.1: A spacetime diagram. Time runs up the page. A particle at rest has a vertical worldline; a moving particle has a tilted one. With $c = 1$, light travels along lines at 45° .

2.2 The interval

In ordinary three-dimensional space, two observers who use differently oriented axes disagree about the coordinates of two points but agree on the distance between them. The squared distance $\Delta x^2 + \Delta y^2 + \Delta z^2$ is the same in every set of rotated axes.

Special relativity has an exact analogue. Observers moving relative to one another disagree about time differences and about distances, yet all inertial observers agree on the *interval*

$$\Delta s^2 = -\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2. \quad (2.1)$$

The minus sign in front of the time difference is the whole difference between the geometry of spacetime and the geometry of space. It divides pairs of events into three kinds.

Definition 2.1 (Kinds of interval). The interval between two events is

- *timelike* if $\Delta s^2 < 0$: a clock can travel from one event to the other at less than the speed of light;
- *null* (or *lightlike*) if $\Delta s^2 = 0$: a light signal can travel from one to the other;
- *spacelike* if $\Delta s^2 > 0$: nothing, not even light, can travel from one to the other.

The events that light can reach from an event A form a cone, the *future light cone* of A (Figure 2.2). Events inside it can be influenced by what happens at A ; events outside it cannot. The events from which light can reach A form its past light cone. The light cones of all events together are the *causal structure* of spacetime: they tell you what can affect what. Every question about faster-than-light travel is, at bottom, a question about light cones.

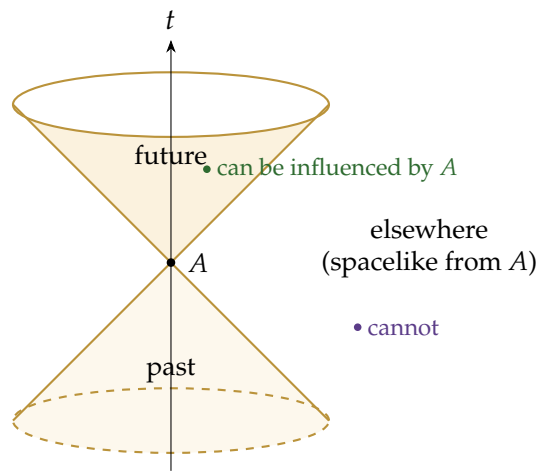


Figure 2.2: The light cone of an event A , drawn with one dimension of space suppressed. Events inside the future cone can be influenced by A . Events outside both cones are spacelike from A , and no signal connects them to it.

Check your understanding 2.1

Event B happens 3 s after event A and 4 light-seconds away from it. Event C happens 5 s after A and 4 light-seconds away. Classify the intervals AB and AC . Can anything that happens at A affect B ? Can it affect C ?

2.3 Proper time

A clock carried along a worldline measures the *proper time* τ along it. For a clock at rest, proper time is just coordinate time. For a moving clock, the interval does the bookkeeping. Along a short piece of a timelike worldline the interval is negative, and the clock records

$$d\tau^2 = -ds^2 = dt^2 - dx^2 - dy^2 - dz^2. \quad (2.2)$$

Dividing by dt^2 gives $d\tau/dt = \sqrt{1 - v^2}$, where v is the clock's speed. The proper time along a whole worldline is the sum of these pieces:

$$\tau = \int \sqrt{1 - v(t)^2} dt. \quad (2.3)$$

This is the time dilation of special relativity written as a geometric length: proper time is to a worldline what length is to a curve in space.

Worked Example 2.1 A trip to the nearest star

Proxima Centauri is about 4.24 light-years away. A traveller leaves Earth at $v = 0.8$, reaches the star, turns around at once, and returns at the same speed. How much time passes on Earth, and on the traveller's clock?

Solution. In the Earth's frame the trip covers $2 \times 4.24 = 8.48$ light-years at speed 0.8, so it takes

$$\Delta t = \frac{8.48}{0.8} \text{ years} = 10.6 \text{ years}.$$

The traveller moves at constant speed on each leg, so Equation (2.3) gives

$$\Delta\tau = \sqrt{1 - 0.8^2} \Delta t = 0.6 \times 10.6 \text{ years} = 6.36 \text{ years}.$$

What it tells us. The traveller returns 4.2 years younger than a twin who stayed home. The trip is not symmetric: the Earth twin's worldline is straight, while the traveller's worldline has a corner at the turnaround. In the geometry of spacetime, the straight worldline between two events is the one with the *longest* proper time, which is the reverse of ordinary geometry, where the straight line is the shortest. The minus sign in Equation (2.1) is responsible.

Key idea

Between two events, the straight worldline of a freely moving clock records the longest proper time. Every detour, every acceleration, shortens the time a clock records.

2.4 The metric

Equation (2.1) holds for inertial observers using ordinary Cartesian coordinates. We need a version that works in any coordinates, and later in curved spacetime. The idea is to write down the interval between *neighbouring* events, as a formula in whatever coordinates we are using.

2.4.1 Index notation

We label the four coordinates with an index that runs from 0 to 3: $x^0 = t$, $x^1 = x$, $x^2 = y$, $x^3 = z$. The raised index is a label, not a power. Greek letters $\mu, \nu, \dots$ stand for any of the four values. The interval between neighbouring events can then be written

$$ds^2 = \sum_{\mu=0}^3 \sum_{\nu=0}^3 g_{\mu\nu} dx^\mu dx^\nu \equiv g_{\mu\nu} dx^\mu dx^\nu. \quad (2.4)$$

The sixteen numbers $g_{\mu\nu}$ form a symmetric 4×4 matrix called the *metric*. In the second form we have used the *summation convention*: an index that appears once up and once down in a term is summed from 0 to 3. We use the convention from now on.

For flat spacetime in Cartesian coordinates, comparing with Equation (2.1) gives the Minkowski metric

$$g_{\mu\nu} = \eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.5)$$

2.4.2 Coordinates are labels

The same flat spacetime looks different in other coordinates. In spherical coordinates the interval reads

$$ds^2 = -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (2.6)$$

The metric components now depend on position, yet nothing physical has changed. Coordinates are labels we attach to events, and a good deal of the skill of working with metrics lies in telling which features of a metric belong to the labels and which belong to the geometry. The next example shows how dramatic a pure coordinate effect can look.

Worked Example 2.2 Flowing coordinates

Consider the metric

$$ds^2 = -dt^2 + (dx - v_0 dt)^2 + dy^2 + dz^2, \quad (2.7)$$

with v_0 a constant. Find the coordinate speed dx/dt of light moving along x . Then show that this metric describes flat spacetime.

Solution. Light has $ds^2 = 0$. Along x we set $dy = dz = 0$ and find $(dx - v_0 dt)^2 = dt^2$, so $dx - v_0 dt = \pm dt$ and

$$\frac{dx}{dt} = v_0 \pm 1.$$

For $v_0 = 0.5$, light moves at $+1.5$ in one direction and -0.5 in the other, as measured in these coordinates.

Now change coordinates to $x' = x - v_0 t$, keeping t, y and z . Then $dx' = dx - v_0 dt$, and the metric becomes $ds^2 = -dt^2 + dx'^2 + dy^2 + dz^2$, the Minkowski metric.

What it tells us. This is flat spacetime described by an observer who labels points with a grid that slides along x at speed v_0 relative to inertial observers. In the sliding grid, light seems to move faster than 1 in one direction. No object ever outruns a light signal sent beside it: the speeds $v_0 \pm 1$ describe how the labels move, not how anything moves relative to its surroundings. Keep this example in mind. The metrics of warp drives have exactly this form, with v_0 replaced by a function that varies from place to place, and in that case the effect is no longer only a matter of labels.

The form Equation (2.7) gives us a picture that will run through this book: space *flowing* past a grid of coordinates. Observers who are carried along by the flow feel no force. The flow speed can take any value, and where it varies from place to place the geometry is genuinely curved.

Check your understanding 2.2

In Worked Example 2.2, suppose $v_0 = 2$. In which direction can light move, as measured in the sliding coordinates? Does this mean something is moving faster than light?

2.4.3 Four-velocity

The tangent to a worldline, measured per unit proper time, is the *four-velocity* $u^\mu = dx^\mu/d\tau$. Dividing Equation (2.4) by $d\tau^2 = -ds^2$ gives the normalization

$$g_{\mu\nu} u^\mu u^\nu = -1. \quad (2.8)$$

For a clock at rest in flat spacetime, $u^\mu = (1, 0, 0, 0)$. For a clock moving at speed v along x , $u^\mu = \gamma(1, v, 0, 0)$ with $\gamma = 1/\sqrt{1-v^2}$. We will use four-velocities in Chapter 3 to say what an observer measures.

2.5 Clocks in curved spacetime

In curved spacetime no choice of coordinates puts the metric into the Minkowski form everywhere at once. The metric still gives the interval between neighbouring events through Equation (2.4), and the proper time along a worldline is still $\tau = \int \sqrt{-ds^2}$. Everything we did above carries over; only the metric changes.

2.5.1 Weak gravity

Around a planet or a star the gravitational field is weak, and the metric is very close to Minkowski. To an excellent approximation,

$$ds^2 = -(1 + 2\Phi) dt^2 + (1 - 2\Phi)(dx^2 + dy^2 + dz^2), \quad (2.9)$$

where Φ is the Newtonian gravitational potential divided by c^2 . For a spherical mass M , $\Phi = -GM/(c^2 r)$, which is small and negative: $\Phi \approx -7 \times 10^{-10}$ at the Earth's surface. The approximation holds whenever $|\Phi| \ll 1$.

A clock moving slowly at speed v through this spacetime records, from Equation (2.9),

$$\frac{d\tau}{dt} = \sqrt{1 + 2\Phi - (1 - 2\Phi)v^2} \approx 1 + \Phi - \frac{1}{2}v^2, \quad (2.10)$$

keeping only the leading terms in the small quantities Φ and v^2 . The two corrections have a clear meaning. A clock deeper in the potential (more negative Φ) runs slower, and a moving clock runs slower.

Worked Example 2.3 Clocks on the ground and in orbit

Navigation satellites orbit at a radius $r_s = 26\,560$ km from the Earth's centre. Take $GM = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$ for the Earth and $R = 6371$ km for its radius, and ignore the Earth's rotation. By how much does a satellite clock gain or lose each day, compared with a clock on the ground?

Solution. The ground clock is at rest at $r = R$, so its rate is $1 + \Phi(R)$. The satellite clock moves on a circular orbit at speed $v = \sqrt{GM/r_s}$, about 3.87 km/s. The difference in rates is

$$\frac{d\tau_s}{dt} - \frac{d\tau_g}{dt} = \underbrace{\frac{GM}{c^2} \left(\frac{1}{R} - \frac{1}{r_s} \right)}_{\text{altitude}} - \underbrace{\frac{v^2}{2c^2}}_{\text{speed}}.$$

Numerically the altitude term is 5.29×10^{-10} and the speed term is 8.35×10^{-11} . Over one day, 86 400 s, the satellite clock gains 45.7 μs from its altitude and loses 7.2 μs from its speed, a net gain of about 38.5 μs per day.

What it tells us. Light travels 11.5 km in 38.5 μs , so an uncorrected clock would spoil positions by kilometres within a day. The full treatment, including the Earth's rotation and shape, is given by Ashby (2003). This is the calculation behind the numbers quoted in Section 1.1.

2.5.2 Near a black hole

For a non-rotating mass M with no weak-field approximation at all, the geometry outside the mass is described by the *Schwarzschild metric*:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) dt^2 + \frac{dr^2}{1 - r_s/r} + r^2(d\theta^2 + \sin^2 \theta d\phi^2), \quad r_s = \frac{2GM}{c^2}. \quad (2.11)$$

Here r is defined so that a sphere at radius r has area $4\pi r^2$. A clock held at fixed r records $d\tau/dt = \sqrt{1 - r_s/r}$ relative to a clock far away. For the Sun, $r_s \approx 2.95 \text{ km}$, so the effect at its surface is tiny. For a neutron star of 1.4 solar masses and radius 12 km, the surface clock runs at 0.81 of the rate of a distant clock (Problem 2.5).

At $r = r_s$ the clock rate would fall to zero. No clock can be held at rest there: $r = r_s$ is the *event horizon* of a black hole. The metric Equation (2.11) also misbehaves at the horizon, where g_{rr} becomes infinite. That part is a coordinate effect, as another way of writing the same geometry shows.

2.5.3 Space as a flowing river

Introduce a new time coordinate T , the time on the clocks of observers who fall inward from rest far away. In these coordinates the Schwarzschild geometry takes the Painlevé–Gullstrand form (in units with $c = 1$):

$$ds^2 = -dT^2 + \left(dr + \sqrt{\frac{r_s}{r}} dT\right)^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2). \quad (2.12)$$

Compare this with Equation (2.7). It has the same form, with space flowing *inward* at the speed $\sqrt{r_s/r}$, which grows as r shrinks. Nothing is singular at $r = r_s$ any more.

Worked Example 2.4 Where light stands still

Use Equation (2.12) to find the coordinate speed dr/dT of light moving radially, and locate the place where outgoing light makes no progress.

Solution. For radial light, $ds^2 = 0$ with $d\theta = d\phi = 0$, so $dr + \sqrt{r_s/r} dT = \pm dT$, or

$$\frac{dr}{dT} = -\sqrt{\frac{r_s}{r}} \pm 1.$$

Ingoing light (− sign) always moves inward. Outgoing light (+ sign) moves outward only where $\sqrt{r_s/r} < 1$, that is, where $r > r_s$. At $r = r_s$ it stands still, and inside it is swept inward.

What it tells us. This is the horizon, seen as a place where space flows inward faster than light can move through it. Light always moves at speed 1 relative to the space around it; the horizon forms where the flow itself exceeds 1. A warp drive uses the same mechanism in reverse, carrying a region along by a flow of space. It also inherits the same danger: wherever the flow outruns light, horizons can form. You will meet those horizons again in Part III.



Figure 2.3: The Schwarzschild geometry pictured as space flowing inward (Equation (2.12)).

Top: the inward flow speed $\sqrt{r_s/r}$ at six radii, drawn to scale. Bottom: the net coordinate velocity $1 - \sqrt{r_s/r}$ of outgoing light at the same radii. Outside the horizon light makes headway; at the horizon it stands still; inside, the flow sweeps it inward.

2.6 Free fall and geodesics

In Worked Example 2.1 we found that the straight worldline between two events has the longest proper time. General relativity turns this into the law of motion for anything on which no non-gravitational force acts:

Key idea: free fall

A freely falling body follows a worldline of extremal proper time, a *geodesic* of spacetime. Gravity is not a force that deflects it; it is the curvature that bends what “straight” means.

Making the proper time extremal leads, after a calculation you can do in Problem 2.8, to the *geodesic equation*

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (2.13)$$

where the *Christoffel symbols* are built from first derivatives of the metric:

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2} g^{\mu\nu} (\partial_\alpha g_{\nu\beta} + \partial_\beta g_{\nu\alpha} - \partial_\nu g_{\alpha\beta}). \quad (2.14)$$

Here ∂_α means $\partial/\partial x^\alpha$, and $g^{\mu\nu}$ is the inverse of the matrix $g_{\mu\nu}$. In flat spacetime with Cartesian coordinates every Γ vanishes, and Equation (2.13) says that free bodies move in straight lines at constant speed.

Worked Example 2.5 Newton’s law of gravity from geometry

Show that for a slowly moving body in the weak-field metric Equation (2.9), the geodesic equation reduces to Newton’s law $d^2 x^i/dt^2 = -\partial_i \Phi$, for a potential that does not change in time.

Solution. A slow body has $dx^i/d\tau \ll dt/d\tau$, so in Equation (2.13) the term with $\alpha = \beta = 0$ dominates:

$$\frac{d^2 x^i}{d\tau^2} \approx -\Gamma^i_{00} \left(\frac{dt}{d\tau} \right)^2.$$

With $g_{00} = -(1 + 2\Phi)$ independent of time, Equation (2.14) gives $\Gamma^i_{00} = -\frac{1}{2} g^{ii} \partial_i g_{00} \approx \partial_i \Phi$ to first order. Since $dt/d\tau \approx 1$ for a slow body,

$$\frac{d^2 x^i}{dt^2} \approx -\partial_i \Phi.$$

What it tells us. Newton’s law of gravity emerges from the tt part of the metric alone, that is, from the way the rate of clocks changes from place to place. Objects fall toward regions where clocks run slower. This is one of the most important intuitions in the book: the rate of clocks, and how it varies across space, is itself a gravitational field.

2.7 Curvature and tides

A single freely falling observer feels nothing: in a small enough laboratory, free fall is indistinguishable from floating in empty space. This is the *equivalence principle*, and it is why the metric can always be made to look like Minkowski at any one event by a suitable choice of coordinates. Gravity reveals itself when you compare *nearby* free-fall paths.

Drop two balls side by side above the Earth. Both fall toward the Earth's centre, so their paths converge and the balls drift together. Drop two balls one above the other, and the lower one, closer to the Earth, accelerates slightly more; they drift apart. These relative accelerations are *tides*, and no choice of coordinates can remove them. They are the true signature of curvature.

2.7.1 The Riemann tensor

In general relativity the relative acceleration of two nearby free-fall paths separated by a small displacement ξ^μ is

$$\frac{d^2 \xi^\mu}{d\tau^2} = -R^\mu{}_{\alpha\nu\beta} u^\alpha \xi^\nu u^\beta, \quad (2.15)$$

where u^μ is the four-velocity of the falling bodies and $R^\mu{}_{\alpha\nu\beta}$ is the *Riemann curvature tensor*:

$$R^\mu{}_{\alpha\nu\beta} = \partial_\nu \Gamma^\mu{}_{\alpha\beta} - \partial_\beta \Gamma^\mu{}_{\alpha\nu} + \Gamma^\mu{}_{\nu\lambda} \Gamma^\lambda{}_{\alpha\beta} - \Gamma^\mu{}_{\beta\lambda} \Gamma^\lambda{}_{\alpha\nu}. \quad (2.16)$$

The Riemann tensor vanishes everywhere if and only if spacetime is flat. When Worked Example 2.2 asked whether a metric describes flat spacetime, computing its Riemann tensor would have answered the question directly, without having to guess the right change of coordinates.

In the weak-field limit, the tidal part of the Riemann tensor becomes the familiar Newtonian tidal tensor, $R^i{}_{0j0} \approx \partial_i \partial_j \Phi$. Near the Earth the tidal acceleration across a person 2 m tall is about $2GML/r^3 \approx 6 \times 10^{-6} \text{ m/s}^2$, far too small to notice. Near a black hole of a few solar masses it would tear a person apart. Tides are one of the first things a spacetime engineer checks on behalf of a traveller.

2.7.2 Ricci and Einstein

Two combinations of the Riemann tensor carry the information Einstein's equation needs. The *Ricci tensor* and the *Ricci scalar* are

$$R_{\alpha\beta} = R^\mu{}_{\alpha\mu\beta}, \quad R = g^{\alpha\beta} R_{\alpha\beta}, \quad (2.17)$$

and the *Einstein tensor* is

$$G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2} R g_{\alpha\beta}. \quad (2.18)$$

The Einstein tensor is the “curvature of spacetime” on the left side of Einstein’s equation. Chapter 3 explains why this particular combination appears, and uses it to compute what a geometry demands.

Going deeper: tensors and coordinate independence

The objects in this chapter are *tensors*: quantities whose components change in a definite way under a change of coordinates, so that equations between tensors hold in every coordinate system if they hold in one. The metric is a symmetric tensor of type $(0, 2)$; the Riemann tensor has type $(1, 3)$. The Christoffel symbols are *not* tensor components, which is why they can vanish at a point in one set of coordinates and not in another. Spacetime is modelled as a four-dimensional Lorentzian manifold, and all the curvature quantities here follow the sign conventions of Misner et al. (1973). Wald’s text develops the same material in coordinate-free, abstract-index form (Wald 1984).

Summary

- With $c = 1$, time and distance share units and light moves at 45° on a spacetime diagram.
- The interval $\Delta s^2 = -\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2$ is the same for all inertial observers. It is timelike, null or spacelike, and light cones divide spacetime into what an event can and cannot influence.
- A clock measures proper time $\tau = \int \sqrt{-ds^2}$ along its worldline. Between two events, the free-fall worldline records the longest proper time.
- The metric $g_{\mu\nu}$ gives the interval between neighbouring events in any coordinates. Coordinate effects can be dramatic, as in flowing coordinates, where light’s coordinate speed differs from 1.
- In a weak field a clock runs at $d\tau/dt \approx 1 + \Phi - v^2/2$. The GPS satellites gain about $38 \mu\text{s}$ per day for this reason.
- The Schwarzschild geometry can be pictured as space flowing inward; its horizon is where the flow reaches the speed of light.
- Free bodies follow geodesics, and the rate of clocks from place to place acts as a gravitational field. The Riemann tensor measures tides, the true signature of curvature; the Ricci and Einstein tensors are built from it.

Answers to Check your understanding

- 2.1 $AB: \Delta s^2 = -9 + 16 = +7$, spacelike, so nothing at A can affect B . $AC: \Delta s^2 = -25 + 16 = -9$, timelike, so a signal slower than light can carry an influence from A to C .
- 2.2 With $v_0 = 2$, both solutions $dx/dt = 2 \pm 1$ are positive: light moves only toward $+x$ in the sliding coordinates. Nothing outruns light. The grid slides at twice the speed of light relative to inertial observers, which is allowed because a grid of labels is not an object.

Problems

- 2.1 [★] Two events are separated by $\Delta t = 2$ s and $\Delta x = 1.5$ light-seconds. Find the interval and classify it. What proper time does a clock record if it travels between them at constant velocity?
- 2.2 [★] A clock travels at $v = 0.99$ for one year of Earth time. How much time does it record?
- 2.3 [★★] **Accelerating twin.** Repeat Worked Example 2.1 with a more realistic trip: the traveller accelerates uniformly from rest to 0.8 over the first year of Earth time, cruises, and decelerates symmetrically at the star and on the way home. Set up the proper-time integral and evaluate it numerically.
- 2.4 [★] **Everest.** By what fraction does a clock on the summit of Mount Everest (8.85 km) run faster than one at sea level? How many nanoseconds does it gain per year?
- 2.5 [★★] **Neutron star.** Show that a clock on the surface of a neutron star of mass $1.4 M_\odot$ and radius 12 km runs at 0.81 of the rate of a distant clock. Use $GM_\odot/c^2 = 1.477$ km.
- 2.6 [★★] **Polar coordinates.** For the flat two-dimensional metric $ds^2 = dr^2 + r^2 d\phi^2$, compute all the Christoffel symbols from Equation (2.14). Show that the geodesic equation has straight lines as solutions.
- 2.7 [★★] **Flowing coordinates, again.** Compute the Christoffel symbols of the metric Equation (2.7) and show that the Riemann tensor vanishes. Then repeat with v_0 replaced by a function $v(y)$, and show that the Riemann tensor no longer vanishes.
- 2.8 [★★] **Deriving the geodesic equation.** Write the proper time along a worldline as $\tau = \int \sqrt{-g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} d\lambda$, where the dot is $d/d\lambda$. Apply the Euler-Lagrange equations and choose $\lambda = \tau$ to obtain Equation (2.13).
- 2.9 [★★] **The river.** Show that Equation (2.12) turns into Equation (2.11) under the change of time coordinate $dT = dt + \frac{\sqrt{r_s/r}}{1-r_s/r} dr$.

- 2.10 [★ ★ ★] **Tides near a black hole.** Estimate the stretching acceleration across a 2 m tall person falling feet first into a black hole of $10 M_{\odot}$ at the moment they cross the horizon, and again for a black hole of $10^9 M_{\odot}$. Which one could a traveller cross unharmed, and why does the answer depend on the mass the way it does?

Further reading

Hartle (2003)

A physics-first introduction to general relativity that builds intuition for metrics, clocks and geodesics before the formalism.

Schutz (2009)

A careful, gentle development of spacetime diagrams, tensors and curvature, well suited to self-study.

Carroll (2004)

A graduate-level introduction that covers the geometry of this chapter with full mathematical care.

Misner et al. (1973)

The encyclopedic classic, with deep geometric intuition; its conventions are the ones used in this book.

Chapter 3

How Matter Shapes Spacetime

What you will learn

- Describe the energy, momentum and stress of matter with the stress-energy tensor, and say what each of its components means.
- Compute the energy density that any observer measures.
- Write down the stress-energy of dust, fluids, radiation, vacuum energy and tension, and explain why tension counts as negative pressure.
- State Einstein's equation, convert between geometric and everyday units, and recover Newton's law of gravity from it.
- Carry out the spacetime engineer's central calculation: from a chosen metric to the matter it demands.
- Recognize the first signs of an exotic demand.

Chapter 2 gave us the left side of Einstein's equation, the geometry. This chapter supplies the right side, the description of matter, and then puts the two together. By the end you will be able to take any metric, compute the matter it requires, and read the result the way a spacetime engineer does: as a list of demands that some physical material would have to meet.

3.1 Energy depends on who measures it

In Newtonian physics, the mass of a brick is a fixed number. In relativity the brick's *energy* depends on the observer: someone moving past it sees it carry kinetic energy as well as its rest energy. To describe matter in a way all observers can use, we need to say how the measured energy changes with the observer.

A particle of mass m with four-velocity u^μ has four-momentum $p^\mu = m u^\mu$. An observer with four-velocity w^μ measures the particle's energy to be

$$E = -g_{\mu\nu} p^\mu w^\nu \equiv -p_\mu w^\mu. \quad (3.1)$$

In the second form we have *lowered an index*: $p_\mu = g_{\mu\nu} p^\nu$. Lowering indices with the metric is the bookkeeping that turns a pair of vectors into a number every observer agrees on. For an observer at rest in flat spacetime, $w^\mu = (1, 0, 0, 0)$, and Equation (3.1) gives $E = p^0$, the time component of the four-momentum. For a particle at rest relative to the observer, $E = m$: its rest energy, mc^2 in ordinary units.

3.2 The stress-energy tensor

A single particle is described by one four-momentum. A continuous material, such as a gas, a fluid or a stretched rubber band, needs more. At each event we want to know how much energy is present per unit volume, how fast energy is flowing, how much momentum there is, and what forces neighbouring parts of the material exert on each other. All of this fits into one symmetric 4×4 array, the *stress-energy tensor* $T^{\mu\nu}$.

Definition 3.1 (Stress-energy tensor). In the frame of an observer, with coordinates t, x, y, z ,

- T^{tt} is the *energy density*, the energy per unit volume;
- $T^{ti} = T^{it}$ is the *energy flux* in direction i , which equals the density of momentum in direction i ;
- T^{ij} is the *stress*: the flux of i -momentum across a surface facing the j direction. Its diagonal entries are *pressures*, and its off-diagonal entries are *shear stresses*.

	t	x	y	z	
t					energy density T^{tt}
x					energy flux = momentum density $T^{ti} = T^{it}$
y					stress T^{ij} : pressures on the diagonal, shear off the diagonal
z					

Figure 3.1: The components of the stress-energy tensor in an observer's frame. The array is symmetric, so ten numbers describe the matter at each event.

Different observers split the same tensor into energy, flux and stress differently, just as observers in relative motion split an interval into time and distance differently. The rule that makes this precise generalizes Equation (3.1): an observer with four-velocity w^μ measures the energy density

$$\rho_w = T_{\mu\nu} w^\mu w^\nu, \quad (3.2)$$

where $T_{\mu\nu} = g_{\mu\alpha} g_{\nu\beta} T^{\alpha\beta}$ is the tensor with both indices lowered. This formula is one of the most used in the book. When a geometry demands matter, the first question is always what energy density each observer would measure.

Worked Example 3.1 A moving cloud of dust

A cloud of dust has particles of mass m at rest relative to one another, with n particles per unit volume in their own rest frame. Find the components of $T^{\mu\nu}$ measured by an observer past whom the cloud moves at speed v along x .

Solution. In the cloud's rest frame each particle has energy m , so the energy density is $\rho_0 = nm$, and there is no flux and no stress. The moving observer sees two changes. Each particle has energy γm , with $\gamma = 1/\sqrt{1-v^2}$. The cloud is also length-contracted along x , so the number density rises to γn . The energy density is therefore

$$T^{tt} = \gamma^2 \rho_0.$$

Energy flows along x at speed v , so the energy flux is $T^{tx} = \gamma^2 \rho_0 v$. Momentum in the x direction, $\gamma^2 \rho_0 v$ per unit volume, is carried along x at speed v , so the momentum flux is $T^{xx} = \gamma^2 \rho_0 v^2$. All other components vanish. These are exactly the components of

$$T^{\mu\nu} = \rho_0 u^\mu u^\nu \quad (3.3)$$

with $u^\mu = \gamma(1, v, 0, 0)$.

What it tells us. The energy density of the same dust is ρ_0 for one observer and $\gamma^2 \rho_0$ for another, and the formula $T^{\mu\nu} = \rho_0 u^\mu u^\nu$ captures both. You can check Equation (3.2) on it: an observer moving with the dust has $w^\mu = u^\mu$, and $T_{\mu\nu} u^\mu u^\nu = \rho_0 (u_\mu u^\mu)^2 = \rho_0$.

3.3 A catalogue of matter

The stress-energy tensors of a few kinds of matter recur throughout this book. Each is written below in its own rest frame, where it is simplest.

Perfect fluid. A fluid with energy density ρ and the same pressure p in every direction has

$$T^{\mu\nu} = (\rho + p) u^\mu u^\nu + p g^{\mu\nu}, \quad T_{\text{rest}}^{\mu\nu} = \text{diag}(\rho, p, p, p). \quad (3.4)$$

Dust is a perfect fluid with $p = 0$. Gases, liquids and the interiors of stars are perfect fluids to a good approximation.

Radiation. A gas of light moving in all directions is a perfect fluid with $p = \rho/3$.

Tension. Stretch a rubber band and each part pulls on its neighbours along the band. A pull is a negative pressure, so tension appears in $T^{\mu\nu}$ as a *negative* diagonal stress. A material under tension τ along z has $T^{zz} = -\tau$. Tension is one of the spacetime engineer's most important ingredients, and you will see it again and again in the demands of engineered geometries.

Vacuum energy. Empty space can carry an energy density of its own, the kind cosmologists invoke to explain the accelerating expansion of the universe. Its stress-energy is

$$T_{\mu\nu} = -\rho_{\text{vac}} g_{\mu\nu}, \quad (3.5)$$

which in any frame has energy density ρ_{vac} and pressure $p = -\rho_{\text{vac}}$ in every direction. Every observer measures the same energy density: vacuum energy looks the same to everyone.

Going deeper: the electromagnetic field

In Heaviside–Lorentz units with $c = 1$, a magnetic field of strength B along z has energy density $B^2/2$, a pressure $+B^2/2$ across the field lines ($T^{xx} = T^{yy}$), and a tension $B^2/2$ along them ($T^{zz} = -B^2/2$). Field lines behave like stretched elastic bands pushing sideways on one another. The tension along the field equals the energy density: this is the largest tension allowed by the dominant energy condition, which you will meet in Chapter 5, and magnetic fields reach it exactly.

Check your understanding 3.1

For each of dust, radiation, a perfect fluid with $p > 0$, and vacuum energy, compute $\rho + p$. Which of them has $\rho + p = 0$? (You will see in Chapter 5 that the sign of $\rho + p$ decides whether matter focuses light.)

3.4 Conservation

Energy and momentum are conserved: whatever flows out of a small box must be accounted for by a loss inside it. In flat spacetime this is the statement

$$\partial_\mu T^{\mu\nu} = 0. \quad (3.6)$$

Its $\nu = t$ component is the continuity equation for energy, $\partial_t T^{tt} + \partial_i T^{it} = 0$; its three spatial components say that momentum changes only through the stresses acting on a region. In curved spacetime the partial derivative is replaced by the *covariant* derivative ∇_μ , which adds Christoffel-symbol terms to account for the geometry: $\nabla_\mu T^{\mu\nu} = 0$. Physically, conservation in curved spacetime says that matter exchanges energy and momentum with the gravitational field in a definite way, and the extra terms describe that exchange.

3.5 Einstein's equation

We can now write Einstein's equation precisely:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (3.7)$$

with $G_{\mu\nu}$ the Einstein tensor of Equation (2.18). There are ten equations here, one for each independent component of the two symmetric tensors.

Going deeper: why the Einstein tensor?

Whatever sits on the left side must be consistent with conservation of energy and momentum on the right side. The Einstein tensor has exactly this property: its covariant divergence vanishes identically, $\nabla^\mu G_{\mu\nu} = 0$ for *every* metric, a result called the contracted Bianchi identity. Einstein's equation therefore implies $\nabla^\mu T_{\mu\nu} = 0$ automatically. The Einstein tensor is essentially the only combination of the metric and its first two derivatives with this property, apart from a term proportional to the metric itself, which acts as vacuum energy.

3.5.1 Geometric units

From now on we set $G = 1$ as well as $c = 1$. Einstein's equation then reads

$$G_{\mu\nu} = 8\pi T_{\mu\nu}. \quad (3.8)$$

In these *geometric units* everything is measured in powers of length. A mass M becomes the length GM/c^2 : the Sun is 1.48 km and the Earth 4.4 mm. Curvature has units of one over

length squared, and so, by Equation (3.8), do energy density and stress. To convert back, multiply a geometric energy density or stress, with lengths in metres, by

$$\frac{c^4}{G} = 1.21 \times 10^{44} \text{ N}, \quad (3.9)$$

to get joules per cubic metre, which are the same as pascals. Divide by c^2 to express an energy density as a mass density. This single conversion factor is the “stiffness of spacetime” of Equation (1.2).

3.5.2 Newton’s law of gravity, recovered

Worked Example 3.2 The Newtonian limit

Show that for the weak-field metric Equation (2.9) with a static potential Φ , the tt component of Einstein’s equation reduces to Poisson’s equation of Newtonian gravity, $\nabla^2\Phi = 4\pi\rho$.

Solution. Compute the Einstein tensor of Equation (2.9) keeping only terms of first order in Φ . The Christoffel symbols are first order, so their products in Equation (2.16) are second order and can be dropped. The calculation (Problem 3.6) gives

$$G_{tt} \approx 2 \nabla^2\Phi.$$

For slowly moving matter, the dominant component of the stress-energy is the energy density, $T_{tt} \approx \rho$. Einstein’s equation $G_{tt} = 8\pi T_{tt}$ then gives

$$\nabla^2\Phi = 4\pi\rho,$$

which is Poisson’s equation in units with $G = 1$.

What it tells us. Newtonian gravity is the weak-field, slow-motion limit of Einstein’s equation. The factor 8π in Equation (3.8) is fixed by this requirement.

3.5.3 Pressure gravitates

Einstein’s equation says more than Newton’s. For a perfect fluid at rest, a short calculation from Equation (3.8) and Equation (3.4) (Problem 3.5) shows that the quantity playing the role of ρ in Poisson’s equation is

$$\rho + 3p. \quad (3.10)$$

Pressure is a source of gravity alongside energy. For ordinary matter the pressure is tiny compared with the energy density, which includes the rest energy mc^2 , so Newton never noticed. Tension, however, *weakens* gravity, and enough of it reverses the sign. Vacuum

energy has $\rho + 3p = -2\rho_{\text{vac}}$: it repels. This is why a positive vacuum energy makes the expansion of the universe accelerate, and it is a first hint of what engineered spacetimes can do with tension.

Key idea

In general relativity every component of the stress-energy tensor bends spacetime: energy, momentum and stress alike. Pressure attracts, tension repels, and a spacetime engineer designs with all of them.

3.6 Reading the equation backwards

Here is the spacetime engineer's central calculation, carried out in full. The recipe has five steps.

1. Write down the metric $g_{\mu\nu}$ of the geometry you want.
2. Compute the Christoffel symbols, Equation (2.14).
3. Compute the Riemann tensor Equation (2.16), then the Ricci tensor, the Ricci scalar and the Einstein tensor Equation (2.18).
4. Read off the demanded stress-energy, $T_{\mu\nu} = G_{\mu\nu}/8\pi$.
5. Interpret it: compute the energy density and stresses measured by the observers you care about, using Equation (3.2).

For all but the simplest metrics, steps 2 and 3 are best done with computer algebra. The examples below are simple enough to follow by hand, and each teaches something that the rest of the book builds on.

Worked Example 3.3 Clocks that run at different rates

Suppose we want a region where clocks run at different rates at different places along x , while space itself stays exactly flat. The metric is

$$ds^2 = -\alpha(x)^2 dt^2 + dx^2 + dy^2 + dz^2, \quad (3.11)$$

where $\alpha(x) > 0$ is the rate of a clock at rest at x , relative to coordinate time. Find the matter this geometry demands.

Solution. *Christoffel symbols.* The only non-constant component is $g_{tt} = -\alpha^2$, which depends on x . Writing $\alpha' = d\alpha/dx$, Equation (2.14) gives only two independent non-zero symbols:

$$\Gamma^t_{tx} = \Gamma^t_{xt} = \frac{\alpha'}{\alpha}, \quad \Gamma^x_{tt} = \alpha \alpha'.$$

Curvature. From Equation (2.16), the non-zero components of the Ricci tensor are

$$R_{tt} = \alpha \alpha'', \quad R_{xx} = -\frac{\alpha''}{\alpha},$$

and the Ricci scalar is $R = g^{tt}R_{tt} + g^{xx}R_{xx} = -2\alpha''/\alpha$. *Einstein tensor.* From Equation (2.18),

$$\begin{aligned} G_{tt} &= \alpha \alpha'' - \frac{1}{2} \left(-\frac{2\alpha''}{\alpha} \right) (-\alpha^2) = 0, \\ G_{xx} &= -\frac{\alpha''}{\alpha} - \frac{1}{2} \left(-\frac{2\alpha''}{\alpha} \right) = 0, \\ G_{yy} &= G_{zz} = -\frac{1}{2} \left(-\frac{2\alpha''}{\alpha} \right) = \frac{\alpha''}{\alpha}. \end{aligned}$$

Interpretation. A clock at rest at x has four-velocity $w^\mu = (1/\alpha, 0, 0, 0)$, and by Equation (3.2) it measures energy density $\rho = G_{tt}/(8\pi\alpha^2) = 0$. The stresses are

$$\rho = 0, \quad p_x = 0, \quad p_y = p_z = \frac{\alpha''}{8\pi\alpha}. \quad (3.12)$$

What it tells us. Changing the rate of clocks from place to place, with space kept flat, demands *no energy at all*, only stress, and only in the directions perpendicular to the change. Where α curves upward ($\alpha'' > 0$), the demand is a pressure; where it curves downward ($\alpha'' < 0$), it is a tension. If α is a straight line, $\alpha = 1 + gx$, the demand vanishes entirely: that geometry is flat spacetime described by an observer with constant acceleration g (Problem 3.7).

Worked Example 3.3 raises a fair question. The Earth makes clocks run at different rates at different heights, yet empty space around the Earth contains no stress. The difference is that the Earth's geometry also curves space: the factor $(1 - 2\Phi)$ in front of the spatial part of Equation (2.9). In empty space the curvature of space and the variation of clock rates cancel exactly in Einstein's equation (Problem 3.6). In Worked Example 3.3 we insisted on keeping space flat, and the stress is the price of that choice. Designs make choices like this all the time, and each one has a price that Einstein's equation computes.

The price is steep. Take a clock-rate change of one part in a billion across one metre, so that $\alpha'' \sim 10^{-9} \text{ m}^{-2}$. By Equations (3.9) and (3.12), the stress is about $10^{-9}/(8\pi) \times 1.2 \times 10^{44} \text{ Pa} \approx 5 \times 10^{33} \text{ Pa}$. That is the scale of pressure inside a neutron star, demanded in order to make two clocks a metre apart differ by one part in a billion.

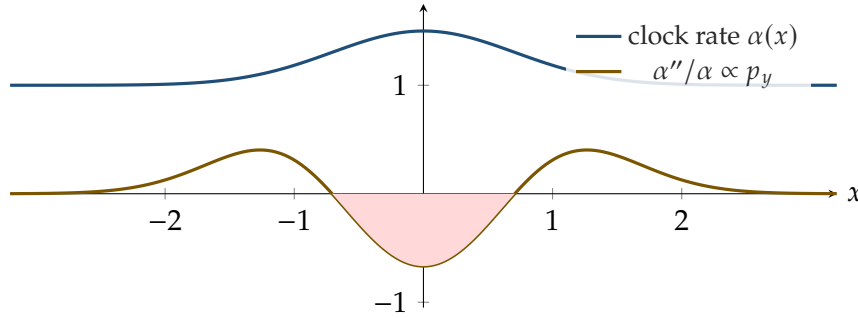


Figure 3.2: A bump in the rate of clocks, $\alpha = 1 + \frac{1}{2}e^{-x^2}$, and the transverse stress it demands (Equation (3.12)). Around the peak the demand is a tension (shaded); on the flanks it is a pressure. The energy density is zero everywhere.

There is a subtler feature too. Consider light moving along y through the peak of the bump, where $p_y < 0$. In Chapter 5 you will learn that ordinary matter always has $\rho + p \geq 0$ for the pressure along any direction light travels. Here $\rho + p_y = \alpha''/(8\pi\alpha) < 0$. A simple bump in the rate of clocks already demands matter that no ordinary material can provide. You will meet this demand again when we study how each part of a design contributes to its cost.

Check your understanding 3.2

In Worked Example 3.3, suppose instead that clocks run *slowest* at $x = 0$, with $\alpha = 1 - \frac{1}{2}e^{-x^2}$. Where is the demand a tension, and where a pressure? Is $\rho + p_y$ negative anywhere?

Worked Example 3.4 A flow of space that shears

Worked Example 2.2 described flat spacetime with coordinates that slide at a constant speed. Now let the speed of the flow vary across space. Consider

$$ds^2 = -dt^2 + (dx + \beta(y) dt)^2 + dy^2 + dz^2, \quad (3.13)$$

a flow of space along x whose speed depends on y . Find the energy density measured by observers who are carried along by the flow.

Solution. Observers carried by the flow have $dx/dt = -\beta$ and feel no force. Their four-velocity is $w^\mu = (1, -\beta, 0, 0)$, which satisfies $g_{\mu\nu}w^\mu w^\nu = -1$ (Problem 3.9). Carrying out the recipe for Equation (3.13) is longer than for Worked Example 3.3, and it is a

good exercise for computer algebra (Problem 3.9). The result for the energy density these observers measure is

$$\rho = T_{\mu\nu}w^\mu w^\nu = -\frac{1}{32\pi} \left(\frac{d\beta}{dy} \right)^2. \quad (3.14)$$

What it tells us. Where the flow is uniform, $d\beta/dy = 0$ and nothing is demanded, just as in Worked Example 2.2. Wherever the flow *shears*, the observers riding it measure a *negative* energy density, whichever way the shear goes, since the derivative is squared. In Chapter 1 we met warp drives: bubbles of flat space carried along by a flow of space. At the bubble's wall the flow changes from the speed of the bubble to zero, so the wall is a region of shear, and Equation (3.14) is the reason a warp drive demands negative energy there. Alcubierre's energy density has exactly this structure (Alcubierre 1994).

The numbers are once again enormous. A flow that changes from zero to the speed of light across one metre has $d\beta/dy \approx 1 \text{ m}^{-1}$, and Equations (3.9) and (3.14) give an energy density of about $-1.2 \times 10^{42} \text{ J/m}^3$, equivalent to a mass density of about $-1 \times 10^{25} \text{ kg/m}^3$. Nuclear matter, the densest material in nature, has about $+2 \times 10^{17} \text{ kg/m}^3$.

3.7 Every geometry has a price

The recipe of Section 3.6 never fails. Every metric you can write down has an Einstein tensor, so every geometry has a demanded stress-energy. Einstein's equation, read backwards, is a price list: it tells you exactly what matter any geometry would require.

Whether anything in nature can pay that price is a separate question, and it is the question the rest of this book is organized around. The examples in this chapter already show its shape. Shaping the rate of clocks demands stress without energy, and can demand matter that violates $\rho + p \geq 0$. Shearing the flow of space demands negative energy density. Chapter 4 splits spacetime into space and time in the way these examples suggest, with a *lapse* that sets the rate of clocks and a *shift* that describes the flow of space, and shows how to compute demands like these efficiently. Chapter 5 states precisely which demands ordinary matter can meet, and Chapter 6 asks how far quantum physics can go beyond them.

Summary

- Energy depends on the observer. An observer with four-velocity w^μ measures the energy density $\rho_w = T_{\mu\nu}w^\mu w^\nu$.

- The stress-energy tensor $T^{\mu\nu}$ collects energy density, energy flux, momentum density and stress into one symmetric array.
- Dust, perfect fluids, radiation, tension and vacuum energy each have a characteristic stress-energy. Tension is negative pressure.
- Einstein's equation $G_{\mu\nu} = 8\pi T_{\mu\nu}$ (with $G = c = 1$) links curvature to matter. Multiply geometric stresses by $c^4/G = 1.21 \times 10^{44}$ N to get pascals.
- Newton's law of gravity is the weak-field limit. Beyond it, pressure gravitates through $\rho + 3p$, and tension can make gravity repel.
- Reading the equation backwards gives the matter any geometry demands. Varying clock rates across flat space demands stress without energy; a shearing flow of space demands negative energy density.

Answers to Check your understanding

- 3.1 Dust: $\rho + p = \rho$. Radiation: $\rho + p = \frac{4}{3}\rho$. Fluid with $p > 0$: $\rho + p > \rho$. Vacuum energy: $\rho + p = 0$. Only vacuum energy sits exactly on the boundary.
- 3.2 Now $\alpha'' > 0$ near $x = 0$, so the demand there is a pressure, and on the flanks (where α curves back toward 1 from below) it is a tension. Wherever the demand is a tension, $\rho + p_y = \alpha''/(8\pi\alpha) < 0$, so it is negative on the flanks.

Problems

- 3.1 [★] A cloud of dust with rest-frame density ρ_0 moves at speed v along y . Write down all components of $T^{\mu\nu}$ in the observer's frame.
- 3.2 [★] **Geometric units.** Express the mass of the Earth, the Sun and a $10^9 M_\odot$ black hole in metres. What energy density, in joules per cubic metre, corresponds to a geometric energy density of $1/\text{km}^2$?
- 3.3 [★★] **Magnetic tension.** Using the magnetic stress-energy in the *Going deeper* box of Section 3.3, verify that $\rho + p \geq 0$ for every direction of the pressure, and that the tension along the field lines equals the energy density.
- 3.4 [★★] **Vacuum energy.** Show from Equation (3.5) that every observer measures the same energy density ρ_{vac} , and that the pressure is $-\rho_{\text{vac}}$ in every direction.
- 3.5 [★★] **Pressure gravitates.** Einstein's equation can be rewritten as $R_{\mu\nu} = 8\pi(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu})$, with $T = g^{\mu\nu}T_{\mu\nu}$. For a perfect fluid at rest, show that $R_{tt} = 4\pi(\rho + 3p)$. Given that

- $R_{tt} \approx \nabla^2 \Phi$ in a weak, static field, explain why $\rho + 3p$ plays the role of ρ in Poisson's equation.
- 3.6 [★★] **Weak fields.** Compute the Einstein tensor of Equation (2.9) to first order in Φ . Show that $G_{tt} \approx 2\nabla^2 \Phi$, and that all components vanish where $\nabla^2 \Phi = 0$. Explain why the result differs from Worked Example 3.3.
- 3.7 [★★] **Uniform acceleration.** Show that the metric Equation (3.11) with $\alpha = 1 + gx$ has vanishing Riemann tensor. What does an observer at rest at fixed x feel?
- 3.8 [★★] **A rippled clock rate.** Take $\alpha = 1 + \epsilon \sin kx$ with $0 < \epsilon < 1$ in Worked Example 3.3. Where is the demand a tension? For light moving along y , where is $\rho + p_y$ negative?
- 3.9 [★★★] **The shearing flow.** Using a computer algebra system, compute the Einstein tensor of Equation (3.13) and verify Equation (3.14). Check that $w^\mu = (1, -\beta, 0, 0)$ is a unit timelike vector. Then compute the energy density measured by an observer at rest in the coordinates, $w^\mu = (1, 0, 0, 0)/\sqrt{1 - \beta^2}$, where $|\beta| < 1$, and compare.
- 3.10 [★★★] **Designing a fast-clock region.** You want a region where clocks run twice as fast as clocks far away, with space kept flat and the transition taking place over 10 m. Using Worked Example 3.3, sketch a suitable $\alpha(x)$, locate the tensions and pressures it demands, and estimate their size in pascals.

Further reading

Schutz (2009)

A careful introduction to the stress-energy tensor through dust and perfect fluids.

Carroll (2004)

Einstein's equation, its Newtonian limit and the Bianchi identity at graduate level.

Hartle (2003)

A physical approach to Einstein's equation and to what curvature does.

Morris and Thorne (1988)

The engineer's calculation of this chapter, applied to traversable wormholes.

Alcubierre (1994)

The original warp drive. After Worked Example 3.4 you can follow its energy density calculation.

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- paper, titled “Hyperfast Interstellar Travel in General Relativity”, pp. 4760–4766. doi: 10.1103/PhysRevD.57.4760. arXiv: gr-qc/9511068v6.
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